

Mahesh Rajarapu

Doctor of Philosophy (Ph. D)

Department of Computer Science & Engineering

Indian Institute of Technology, Tirupati

Links: [Home Page](#) | [LinkedIn](#) | [GitHub](#) | [Leetcode](#)

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Location: Tirupati, Andhra Pradesh, India

Academic Qualifications

- **PhD in Computer Science and Engineering** (July 2025 – Present)
[Indian Institute of Technology Tirupati, Tirupati, Andhra Pradesh, India](#)
Current CGPA: 8.64/10.0
- **Integrated Master of Science in Mathematics** (Dec 2020 – June 2025)
[Sardar Vallabhbhai National Institute of Technology, Surat, Gujarat, India](#)
Final CGPA (after 10 semesters): 7.68/10.0
- **Board of Intermediate Education** (June 2018 – Feb 2020)
[Sri Sarada Educational Institutions, Vijayawada, Andhra Pradesh, India](#)
CGPA: 8.82/10.0
- **Board of Secondary Education** (March 2018)
[Bhashyam E.M High School, Bhavanipuram, Andhra Pradesh, India](#)
CGPA: 9.80/10.0

Research Experience

- **Junior Research Fellow and Ph.D. Researcher** (July 2025 – Present)
[Indian Institute of Technology Tirupati, Tirupati, India](#) | Supervisor: [Dr. Kalidas Yeturu](#).
 - Conducting research in heritage-document analysis, structure-aware image segmentation, privacy-preserving document management, and face image quality assessment.
 - Translating research into production systems through full-stack development, Docker-based deployment, institutional hosting, and human-in-the-loop archival workflows.
- **Graduate Research Intern** (Jun 2025 - Present)
[Indian Institute of Management Mumbai, Mumbai, India](#) | PI: [Prof. Manoj K. Tiwari, Director, IIM Mumbai](#).
 - Developed a novel framework using a fine-tuned Large Language Model (LLM) to optimize drone delivery, integrating real-time meteorological data from the Open-Meteo API for dynamic route planning, energy-efficient battery management, and multi-drone coordination, achieving enhanced reliability and efficiency in urban settings like Delhi.
 - Provided an open-source toolkit with datasets and weather-aware modules to support scalable, sustainable drone logistics.
- **KAI-X Summer Research Internship** (Jun 2024 - Aug 2024)
[Korea Advanced Institute of Science and Technology, Daejeon, South Korea](#) | PI: [Prof. Chang-Ock Lee](#).
 - Focused on simulating computed tomography (CT) and developing advanced image reconstruction algorithms, including classical methods like Filtered Back-Projection (FBP) and deep learning techniques.
 - Aimed to enhance image quality and reduce artifacts by integrating deep learning techniques with a deep understanding of CT physics and data acquisition simulation.
- **Summer Internship Program** (May 2024 - Jun 2024)
[National Institute of Technology Calicut, Kozhikode, Kerala, India](#) | PI: [Dr. Pranesh Das](#).
 - Developed the Dynamic Batch Size Adjuster (DBSA) algorithm to dynamically optimize batch sizes during training, enhancing convergence rates and stability in deep learning models.
 - Improved model performance and minimized computational complexity by adjusting batch sizes in real-time using mathematical analysis of training dynamics, including loss functions and gradient norms.
- **Research Internship** (Ongoing)
[Sardar Vallabhbhai National Institute of Technology, SVNIT, Surat](#) | PI: [Dr. Praveen Kumar Chandaliya](#).
 - Working on child face aging using AI to generate synthetic faces through latent space manipulation, with applications in forensics and healthcare.
 - Working on explainable AI algorithms to assess facial image quality, enhancing image processing techniques across various domains.

- **Research Internship Program** (Sept 2023)
Technical University of Munich, Department of Informatics, Virtual – Munich, Germany | PI: Dr. Nassir Navab.
 - Currently working on 3D Ultrasound Compounding for Volume Estimation in Thyroid Diagnostics.
 - Applied PyTorch on Image Computation to develop cutting-edge solutions for improving Multi-model Thyroid Registration.
- **IAS-INSANA Summer Research Fellowship Program** (May 2023 - July 2023)
Indian Institute of Technology Ropar, DoCSE, Ropar, Punjab, India | PI: Dr. Sudarshan Iyengar.
 - Utilized expertise in machine learning & Deep Learning to work on auto image enhancement, employing advanced techniques and models with optimal gamma values.
 - Applied proficiency in TensorFlow, PyTorch, and JAX to develop cutting-edge solutions for improving image quality and visual appeal.
- **Summer Workout in Mathematics & Programming** (June 2022)
TATA Institute of Fundamental Research - CAM, Bengaluru, Karnataka, India.
 - Studied statistical learning techniques for data analysis, regression, classification, and dimensionality reduction, alongside Python programming proficiency.
 - Developed a solid foundation in real analysis and linear algebra for mathematical and computational problem-solving.

M.Sc. Dissertation

- **Dissertation (Completed)** Grade: 10/10
Sardar Vallabhbhai National Institute of Technology, SVNIT, Surat | Supervisor: Dr. Shivam Bajpeyi.
 - Developed neural-network approximation and reconstruction methods by studying how different activation functions affect learning and reconstruction accuracy.
 - Investigated efficient neural-network operator approximations using Gaussian interpolation and related optimization techniques.

Research Publications

- **AKSHARA: Attention-based Knowledge-preserving Segmentation for Heritage Archival Region Analysis** (Submitted; Under Review)
Mahesh Rajarapu, Vamshi Krishna Shanigarapu, Sahasvat R, Kalidas Yeturu
Submitted to CIKM 2026.
 - Develops attention-based, knowledge-preserving text-region segmentation for retrieval and analysis of degraded palm-leaf manuscript folios.
- **PrivView: Reconstructable Views for Privacy-Preserving Management of Text-Bearing Images** (Submitted; Under Review)
Mahesh Rajarapu, Kalidas Yeturu
Submitted to SIGMOD 2027.
 - Introduces OCR-resistant views that prevent unauthorized text extraction while enabling controlled server-side reconstruction for restricted collections.
- **SegVFNet: Learning Skeleton-Guided Vector Fields for Structure-Aware Semantic Segmentation** (Submitted; Under Review)
Mahesh Rajarapu, Kalidas Yeturu
Submitted to WACV 2027, Algorithms Track.
 - Learns skeleton-guided vector fields to preserve topology and improve structural consistency in semantic segmentation.
- **MU-FIQA: Verification-Margin Supervision and Dual-Stream Fusion for Face Image Quality Assessment** (Accepted: BIOSIG 2026)
Mahesh Rajarapu, Praveen Kumar Chandaliya, Vishnu Chalavadi, Kiran Raja
BIOSIG 2026.
 - Combines verification-margin supervision with dual-stream feature fusion to estimate face-image utility for reliable recognition.
- **Application of Gaussian Activated Neural Network in Image Processing** (Submitted to Neurocomputing)
Mahesh Rajarapu, Shivam Bajpeyi
| Department of Mathematics, Sardar Vallabhbhai National Institute of Technology, Surat, Gujarat-395007.

- Focuses on developing Gaussian-activated neural network models for improved image processing techniques.
- Explores the impact of Gaussian interpolation methods on enhancing reconstruction accuracy in image datasets.
- **Dynamic Batch Size Adjuster (DBSA): Enhancing Deep Learning Training Efficiency** (Submitted)
Mahesh Rajarapu, Pranesh Das*, Ebrahim Ghaderpour
| *Machine Learning Laboratory, Department of Computer Science and Engineering, National Institute of Technology, Calicut, India**.
 - Proposes a novel adaptive batch size mechanism to enhance deep learning model convergence.
 - Aims to improve training efficiency and stability for various neural network architectures.
- **Applications and Performance Analysis of Neural Network Operators in Image Processing** (Accepted Book Chapter (Springer))
Mahesh Rajarapu, Shivam Bajpeyi
Accepted in Neural Networks Approximation with Applications in the AI Age, Springer.
 - Investigates the application and performance of neural network operators in enhancing image processing tasks.
 - Analyzes the efficiency and effectiveness of neural network-based methods for image data processing.
- **Computer-Implemented System and Method for Multi-Mode Digital Annotation of Image-Based Textual Records** (Published Patent Application)
Mahesh Rajarapu, Kalidas Yeturu
Indian Patent Application No. 202641075717; filed June 18, 2026; published June 26, 2026.

Key Projects

- **CT Simulation and Image Reconstruction Algorithms** Source Code
Mentor: Prof. Chang-Ock Lee
 - Simulating CT data acquisition and implementing reconstruction algorithms like Filtered Back-Projection (FBP).
 - Exploring deep learning techniques to enhance image quality and reduce artifacts in imaging.
- **Real-time Object Detection with Deep Learning** Source Code
 - Developed a real-time object detection system using advanced deep learning techniques.
 - Leveraged YOLOv8 architecture for accurate detection of objects in video streams.
 - Created a solution for real-time visual assistance across various applications.
- **Image CompressNet using Encoder-Decoder Architecture** Source Code
 - Designed an autoencoder architecture for binary image reconstruction using the Keras Functional API.
 - Developed an encoder-decoder structure with convolutional and upsampling layers to enhance image reconstruction and model performance.
- **Auto Image Enhancement using Optimal Gamma Values** Source Code
 - Applied Power Law Transformations, CLAHE, and Scaling methods for image enhancement.
 - Developed CNN models for automatic image enhancement, improving contrast, brightness, and saturation.
- **Chronic Kidney Disease Prediction using ML Techniques** Source Code
 - Evaluated various machine learning algorithms to predict kidney disease, identifying the most accurate algorithm.
- **Automated Face Recognition and Attendance Management System** Source Code
 - Developed a face recognition system for entry gate operations.
 - Implemented a deep learning model for accurate identification and attendance logging.

Software Tools Developed

- **ShreeAkshara Annotation Platform** [Live Platform](#)
Human-in-the-loop palm-leaf annotation and multilingual transcription platform with automatic region segmentation, multilayer bounding-box editing, validation workflows, and JSON/PDF export. *React.js, Vite, Flask REST API, Node.js, Docker.*
- **ANVESHANA: Searchable PDF Platform** [Live Platform](#)
Secure manuscript retrieval system supporting transliteration-based browsing, full-text search, matched-folio visualization, and controlled access to archival collections. *React.js, Flask REST API, Node.js.*
- **Palm Leaf Denoiser** [Live Platform](#)
Restores degraded manuscript scans, extracts text-bearing regions, and exports cleaned folio images for transcription and archival use. *React.js, Flask REST API, Node.js.*
- **Kali-OMR** [Live Platform](#)
Automated OMR evaluation platform with geometric correction, bubble detection, configurable scoring, manual verification, and CSV/PDF reporting. *Flask REST API, Node.js, Twilio services.*

Technical Skills

- **Programming:** Python, C, JavaScript, TypeScript, MATLAB
- **AI/ML Frameworks:** PyTorch, TensorFlow, JAX, Flax, Keras, Scikit-Learn, XGBoost, LightGBM
- **Computer Vision and Research:** Image Segmentation, Image Reconstruction, Face Image Quality Assessment, OCR and Document Image Analysis
- **Full-Stack Development:** React.js, Vite, HTML, CSS, Flask, Node.js, Express.js, REST APIs
- **MLOps and DevOps:** Docker, Nginx, Linux Server Deployment, Containerization, Application Monitoring and Maintenance
- **Data and Security:** Database Integration, Structured Data Management, Authentication, Role-Based Access Control
- **Developer Tools:** Git, GitHub, API Testing, Debugging, Technical Documentation
- **Other:** L^AT_EX, Excel, Power BI, MS Office

Funding and Fellowships

- **Sponsorship for KAI-X**
 - Received 1.5 million KRW and housing to participate in the KAI-X Summer Research Internship at Korea Advanced Institute of Science and Technology (KAIST), South Korea (Jun 2024 - Aug 2024).
- **IAS-INSANA-SIRFP Fellowship**
 - Received a monthly stipend for two months during the IAS-INSANA-SIRFP Summer Research Fellowship Program at IIT Ropar, funded jointly by the Indian Academy of Sciences (IAS), Indian National Science Academy (INSA), and National Academy of Sciences, India (NASI) (May 2023 - July 2023).
- **Personal Funding for Integrated M.Sc.**
 - Secured complete personal funding for five years of the Integrated M.Sc. in Mathematics at Sardar Vallabhbhai National Institute of Technology, Surat, covering tuition and academic expenses (Dec 2020 – June 2025).
- **SVNIT Alumni Funding**
 - Received alumni funding from SVNIT Surat on two occasions to support academic and research endeavors during the Integrated M.Sc. program.

Scholastic Achievements

- Awarded the prestigious **KAI-X Summer Internship 2024** at **Korea Advanced Institute of Science and Technology (KAIST)**, South Korea, selected as one of 16 interns from a competitive global pool.
- Awarded fellowship to work under Dr. Sudarshan Iyengar, Dept. of CSE, IIT Ropar by **Indian Academy of Sciences along with INSA, NASI – Summer Research Fellowship**.
- Secured rank in the merit list: Top 53,000 among 1 million candidates in **NIT-JEE Mains 2020**.
- Secured 7th rank in **National Talent Search Examination (NTSE)** conducted in Andhra Pradesh.
- Secured Zonal 6th rank in **National Mathematics Olympiad**.

External Short Term Courses

- **MaLICO_N**
 - One Week Online Short Term Course (e-STC) organized by National Institute of Technology, Hamirpur on "Machine Learning and its applications in Information Security, Computer Vision, and Natural Language Processing (MaLICO_N-2023)" during 13-17 March 2023.
- **Advanced and Technical Computing with MATLAB**
 - Participated in the 5 Days Online Faculty Development Programme on Advanced and Technical Computing with MATLAB organized by the Department of Mathematics, School of Engineering, Presidency University, Bengaluru from 25th to 29th July 2023.

Technical Courses/Electives

Data Structures and Algorithms	Artificial Intelligence
Computer Networks	Graph Theory
Ethical Hacking	Natural Language Processing
Block-chain Technology	Computer Vision
Computer Graphics	Database Management Systems

Note: For a comprehensive list of all courses, please [click here](#).

Talks Delivered/Presentations

- About "Low Cost Sanitary Pad Product Manufacturing Machine" in **Indian Institute of Technology, Delhi** ("UNNATI" MAHOTSAV) and this Project/Machine designed under the UNNAT BHARAT ABHIYAN.

Positions of Responsibility

- **Convener, Unnat Bharat Abhiyan**, SVNIT Surat during A.Y. 2022-23.
 - Inspired by the vision of transformational change in rural development processes by leveraging knowledge institutions to help build the architecture of an Inclusive India.
- **Convener, National Mathematics Day (NMD) 2022**, DoMH, SVNIT Surat.
 - Organized 11 events at the national level as part of the celebrations of NMD.
- **Head, Media and Designing Committee**, Academic Affairs Council (Student's Council) during A.Y. 2022-23.
 - Managed media, designed posters, and created videos for all programs conducted under the AAC Student Council.
- **Co-Head, Media and Designing Committee**, General Student Council during 2022-23.
 - Secured the media, designed posters, and created videos for various programs conducted under the General Student Council.
- **Designer**, Departmental Brochure, Handbook, Pramiti.
 - Designed and developed the departmental brochure, student handbook, and Pramiti magazine, ensuring high-quality design and content presentation.

References

Dr. Kalidas Yeturu

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Dr. Jayesh M. Dhodiya (HoD)

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Dr. Praveen K. Chandaliya

Assistant Professor, Department of
Artificial Intelligence, SVNIT
✉ pkc@aid.svnit.ac.in

Languages

Telugu: Bilingual Proficiency

English: Working Proficiency

Hindi: Conversational

"I believe that "Experience" is the word driving our entire world and universe. Every action, movement and endeavor of humanity is ultimately motivated by the desire to gain new experiences."

- Mahesh Rajarapu